

## Ex. No : 21

## Date : Basics of NumPy

### Aim

To understand the basics of NumPy and perform various operations on NumPy arrays.

### Theory

NumPy (Numerical Python) is a Python library used for numerical and mathematical operations. It provides support for multidimensional arrays, matrix operations, and various mathematical functions.

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### 21.1 Program to Create a NumPy Array

#### Program

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print(arr)
```

#### Sample Output

```
[10 20 30 40 50]
```

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### 21.2 Program to Create 1D and 2D NumPy Arrays

#### Program

```
import numpy as np

arr1 = np.array([1, 2, 3, 4])
arr2 = np.array([[1, 2], [3, 4]])

print("1D Array:")
print(arr1)
```

```
print("2D Array:")  
print(arr2)
```

### Sample Output

```
1D Array:  
[1 2 3 4]  
2D Array:  
[[1 2]  
 [3 4]]
```

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## 21.3 Program to Perform Basic Arithmetic Operations on NumPy Arrays

### Program

```
import numpy as np  
  
arr1 = np.array([1, 2, 3])  
arr2 = np.array([4, 5, 6])  
  
print("Addition:", arr1 + arr2)  
print("Subtraction:", arr1 - arr2)  
print("Multiplication:", arr1 * arr2)  
print("Division:", arr1 / arr2)
```

### Sample Output

```
Addition: [5 7 9]  
Subtraction: [-3 -3 -3]  
Multiplication: [ 4 10 18]  
Division: [0.25 0.4  0.5 ]
```

## 21.4 Program to Find Mean, Median, and Standard Deviation of an Array

### Program

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50])

print("Mean =", np.mean(arr))
print("Median =", np.median(arr))
print("Standard Deviation =", np.std(arr))
```

### Sample Output

```
Mean = 30.0
Median = 30.0
Standard Deviation = 14.142135623730951
```

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## 21.5 Program to Reshape a NumPy Array

### Program

```
import numpy as np

arr = np.array([1, 2, 3, 4, 5, 6])

reshaped = arr.reshape(2, 3)

print(reshaped)
```

### Sample Output

```
[[1 2 3]
 [4 5 6]]
```

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## 21.6 Program to Find Maximum and Minimum Values in an Array

### Program

```
import numpy as np

arr = np.array([12, 45, 7, 89, 23])

print("Maximum Value =", np.max(arr))
print("Minimum Value =", np.min(arr))
```

### Sample Output

```
Maximum Value = 89
Minimum Value = 7
```

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## 21.7 Program to Perform Array Slicing in NumPy

### Program

```
import numpy as np

arr = np.array([10, 20, 30, 40, 50, 60])

print(arr[1:5])
```

### Sample Output

```
[20 30 40 50]
```

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## 21.8 Program to Create Arrays Using zeros(), ones(), and arange()

### Program

```
import numpy as np

print("Zeros Array:")
print(np.zeros((2, 3)))
```

```
print("Ones Array:")
print(np.ones((2, 2)))

print("Arange Array:")
print(np.arange(1, 10, 2))
```

### Sample Output

```
Zeros Array:
[[0. 0. 0.]
 [0. 0. 0.]]
Ones Array:
[[1. 1.]
 [1. 1.]]
Arange Array:
[1 3 5 7 9]
```

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## 21.9 Program to Perform Matrix Addition and Multiplication

### Program

```
import numpy as np

arr1 = np.array([[1, 2], [3, 4]])
arr2 = np.array([[5, 6], [7, 8]])

print("Matrix Addition:")
print(arr1 + arr2)

print("Matrix Multiplication:")
print(np.dot(arr1, arr2))
```

### Sample Output

```
Matrix Addition:
[[ 6  8]
 [10 12]]
Matrix Multiplication:
[[19 22]
 [43 50]]
```

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## 21.10 Program to Find the Transpose of a Matrix

### Program

```
import numpy as np

matrix = np.array([[1, 2, 3], [4, 5, 6]])

print("Original Matrix:")
print(matrix)

print("Transpose Matrix:")
print(matrix.T)
```

### Sample Output

```
Original Matrix:
[[1 2 3]
 [4 5 6]]
Transpose Matrix:
[[1 4]
 [2 5]
 [3 6]]
```

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## Result

The programs demonstrating the basics of NumPy were successfully executed.

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## Viva Questions

1. What is NumPy?
  2. What is the difference between a list and a NumPy array?
  3. What is array slicing in NumPy?
  4. What is the use of reshape() in NumPy?
  5. How is matrix multiplication performed in NumPy?
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## Exercises

1. Write a program to create a 3D NumPy array.

2. Write a program to find the sum and average of array elements.
3. Create two matrices and perform subtraction.
4. Write a program to sort elements in a NumPy array.